

FUNDAMENTALS OF MACHINE LEARNING

MULTIVARIATE METHODS

CSCI3320

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Introduction to Machine Learning

Background

- In previous chapter, we discussed parametric approach to “*classification*” and “*regression*”
- Let us now generalize the technique wherein:
 - ▣ We have *multiple features* (*d dimensions*)
 - ▣ Output is either
 - a particular object (*classification*)
 - a function (*regression*)
- Learn the output function from a labeled multivariate sample (*or supervised learning*)

Parametric Methods

In Chapter 3, we wrote posterior probability for class C_i

$$P(C_i|x) = \frac{p(x|C_i)P(C_i)}{p(x)} = \frac{p(x|C_i)P(C_i)}{\sum_{k=1}^K p(x|C_k)P(C_k)}$$

Using discriminant function

$$g_i(x) = p(x|C_i)P(C_i) \quad \text{or} \quad g_i(x) = \log p(x|C_i) + \log P(C_i)$$

In parametric methods, ~~we assume $p(x|C_i)$ follows some probability distributions,~~ so we can use the probability density function of x to represent $p(x|C_i)$. The only question is how to “*estimate*” parameters of that distribution.

Now the probability density function has to be **multivariate**

Multivariate Data

- Multiple measurements are taken from the targeted inputs.
- d columns of inputs/features/attributes: d -variate
- N rows of instances/observations/examples

$$\mathbf{X} = \begin{bmatrix} X_1^1 & X_2^1 & \cdots & X_d^1 \\ X_1^2 & X_2^2 & \cdots & X_d^2 \\ \vdots & \vdots & \cdots & \vdots \\ X_1^N & X_2^N & \cdots & X_d^N \end{bmatrix} \begin{matrix} 1^{st} \text{ training data} \\ 2^{nd} \text{ training data} \\ \vdots \\ N^{th} \text{ training data} \end{matrix}$$

Example of input

Notes on Multivariate Data

- These variables are **correlated** (**example**)
- Several objectives:
 - ▣ **Simplification**: summarize the large data by means of relatively few parameters
 - ▣ **Exploration**: generating hypotheses about the data
 - ▣ **Prediction**: predict value of one variable from values of other variables. If the predicted value is:
 - **Discrete**: Then it is a “multivariate classification”
 - **Continuous**: Then it is a “multivariate regression”

Multivariate Parameters

How to compute this ?

- **mean vector μ** : $E[\mathbf{x}] = \mu = [\mu_1, \dots, \mu_d]^T$

- **Variance of X_i is σ_i^2**

How to compute this ?

- **Covariance of X_i and X_j is**

How to compute this ?

$$\sigma_{ij} \equiv \text{Cov}(X_i, X_j) = E[(X_i - \mu_i)(X_j - \mu_j)] = E[X_i X_j] - \mu_i \mu_j$$

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1d} \\ \sigma_{21} & \sigma_2^2 & \cdots & \sigma_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{d1} & \sigma_{d2} & \cdots & \sigma_d^2 \end{bmatrix}$$

$$\sigma_{ij} = \sigma_{ji},$$

$$\sigma_{ii} = \sigma_i^2$$

physical meaning

- **Correlation:** $\text{Corr}(X_i, X_j) \equiv \rho_{ij} = \frac{\sigma_{ij}}{\sigma_i \sigma_j}$

Parameter Estimation

The **sample** mean vector :

$$\mathbf{m} = \frac{\sum_{t=1}^N \mathbf{x}^t}{N} \text{ with } m_i = \frac{\sum_{t=1}^N x_i^t}{N}, i = 1, \dots, d$$

The **sample** covariance matrix S where:

$$s_i^2 = \frac{\sum_{t=1}^N (x_i^t - m_i)^2}{N} \quad \text{biased estimates}$$

$$s_{ij} = \frac{\sum_{t=1}^N (x_i^t - m_i)(x_j^t - m_j)}{N}$$

The **sample** correlation matrix R where:

$$r_{ij} = \frac{s_{ij}}{s_i s_j}$$

Covariance Matrix Σ

□ σ_{ij} relates feature x_i with x_j

□ To illustrate, consider

$$\mathbf{x} = [x_1; x_2], \text{ then } \Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{bmatrix}$$

□ What is $\mathbf{x}^T \Sigma \mathbf{x} = ?$

□ We have

$$\mathbf{x}^T \Sigma \mathbf{x} = [x_1, x_2] \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= [\sigma_1^2 x_1 + \sigma_{12} x_2, \sigma_{12} x_1 + \sigma_2^2 x_2] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= \sigma_1^2 x_1^2 + 2\sigma_{12} x_1 x_2 + \sigma_2^2 x_2^2$$

□ Generalize to Σ which is a $d \times d$ matrix

Estimation of Missing Values

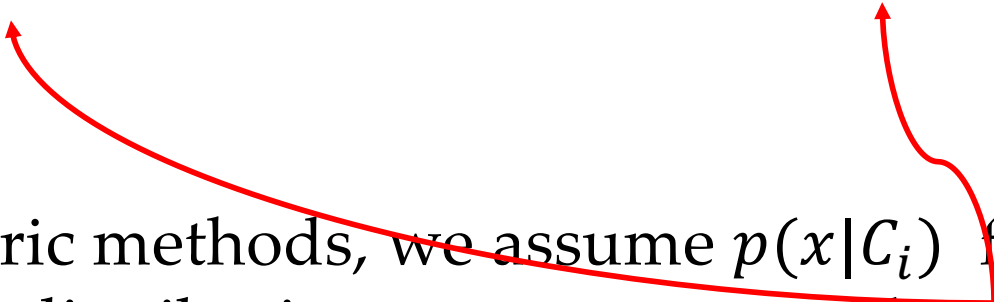
- What to do if certain instances have missing attributes?
- Ignore those instances:
 - ▣ not a good idea if the sample is small
- Use 'missing' as an attribute: may give information
- **Imputation:** Fill in the missing value
 - ▣ **Mean imputation:** Use the most likely value (e.g., mean or most occurred (if discrete))
 - ▣ **Imputation by regression:** Predict based on other attributes

Parametric Methods

Using discriminant function

$$g_i(x) = p(x|C_i)P(C_i) \quad \text{or} \quad g_i(x) = \log p(x|C_i) + \log P(C_i)$$

In parametric methods, we assume $p(x|C_i)$ follows some probability distributions, so we can use the probability density function of x to represent $p(x|C_i)$. The only question is how to “*estimate*” parameters of that distribution.



Now the probability density function has to be **multivariate**

Let's condition ***Multivariate Normal Distribution***

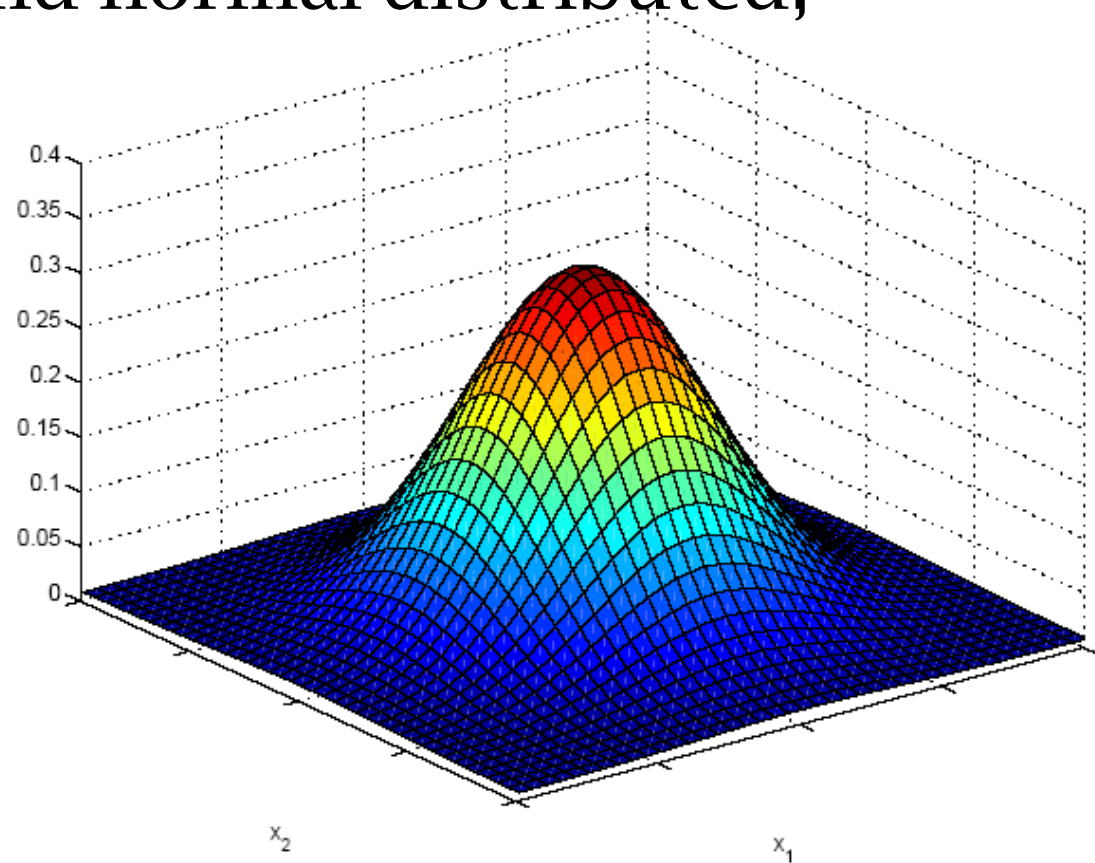
Multivariate Normal Distribution

$\mathbf{x}$ is d -dimensional and normal distributed,

$$\mathbf{x} \sim \mathcal{N}_d(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

$\boldsymbol{\mu}$ is the mean vector

$\boldsymbol{\Sigma}$ is the covariance matrix



$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}|^{1/2}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right]$$



determinant

Matrix inverse

Multivariate Normal Distribution

squared distance from $\mathbf{x}$ to $\boldsymbol{\mu}$ in standard deviation units

$$\frac{(\mathbf{x} - \boldsymbol{\mu})^2}{\sigma^2} = (\mathbf{x} - \boldsymbol{\mu})(\sigma^2)^{-1}(\mathbf{x} - \boldsymbol{\mu})$$

□ **Mahalanobis distance:** $(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})$

Measures the distance from $\mathbf{x}$ to $\boldsymbol{\mu}$ in terms of $\boldsymbol{\Sigma}$ (normalizes for difference in variances and correlations)

□ Bivariate: $d = 2$

$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix} \quad \rho = \frac{\sigma_{12}}{\sigma_1\sigma_2}$$

$$p(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp \left[\frac{-1}{2(1-\rho^2)} (z_1^2 - 2\rho z_1 z_2 + z_2^2) \right]$$

$$\text{where } z_i = \frac{(x_i - \mu_i)}{\sigma_i} \text{ for } i \in \{1, 2\}$$

Multivariate Normal Distribution (Supplement)

□ Determinant

□ If $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $|\mathbf{A}| = ad - bc$

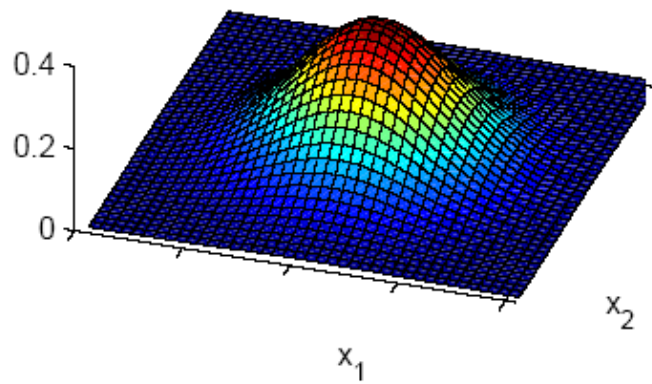
□ If $\mathbf{A} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$, then $|\mathbf{A}| = aei + bfg + cdh - ceg - bdi - afh$

□ Inverse

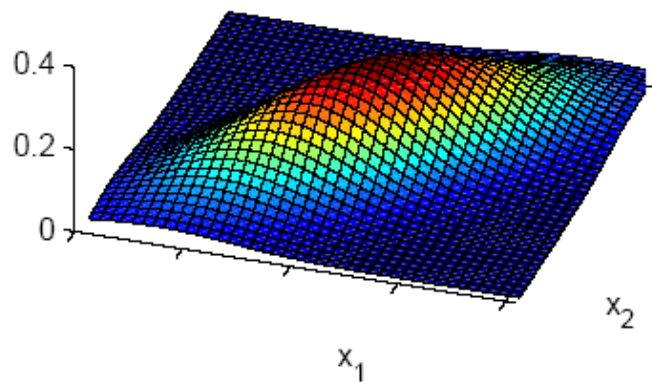
□ If $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $\mathbf{A}^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{|\mathbf{A}|} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

□ For higher dimension, you can do Google lookup

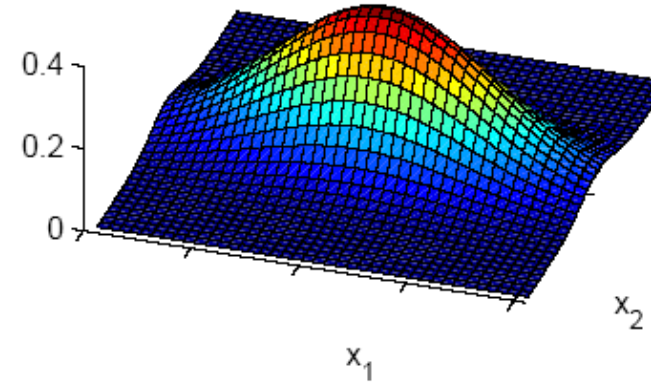
$\text{Cov}(x_1, x_2) = 0, \text{Var}(x_1) = \text{Var}(x_2)$



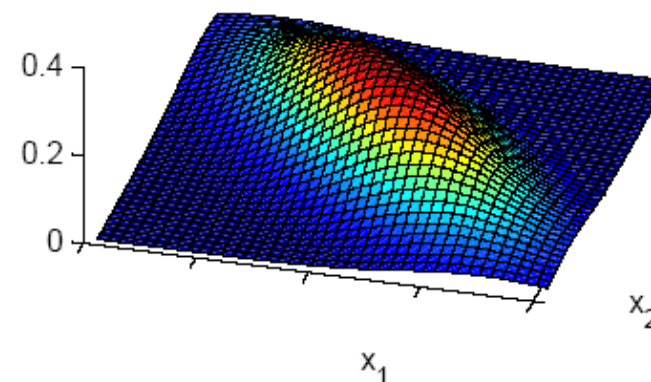
$\text{Cov}(x_1, x_2) > 0$



$\text{Cov}(x_1, x_2) = 0, \text{Var}(x_1) > \text{Var}(x_2)$

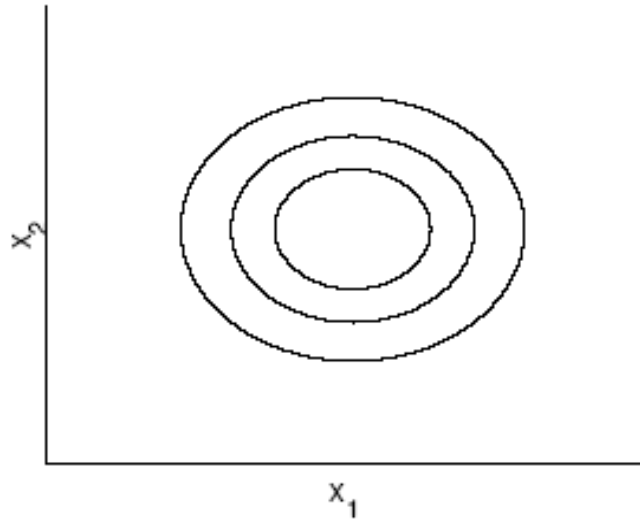


$\text{Cov}(x_1, x_2) < 0$

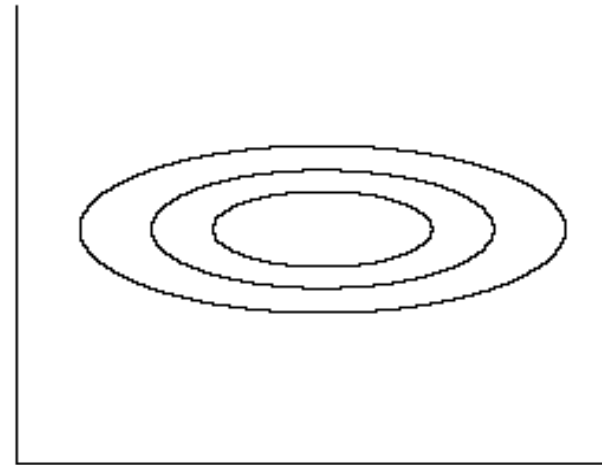


Bivariate Normal, $d=2$, **contour plot**

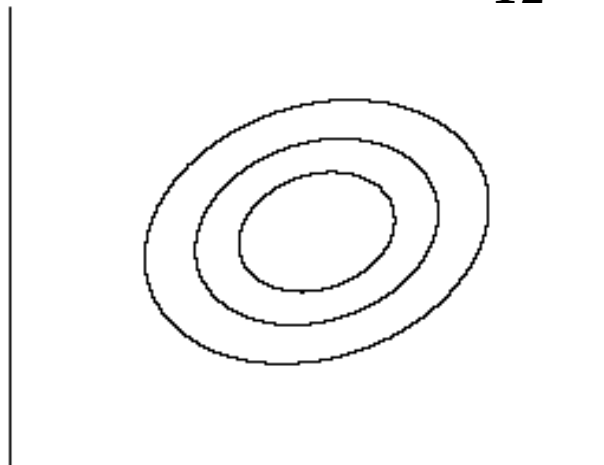
$\text{Cov}(x_1, x_2)=0, \text{Var}(x_1)=\text{Var}(x_2)$



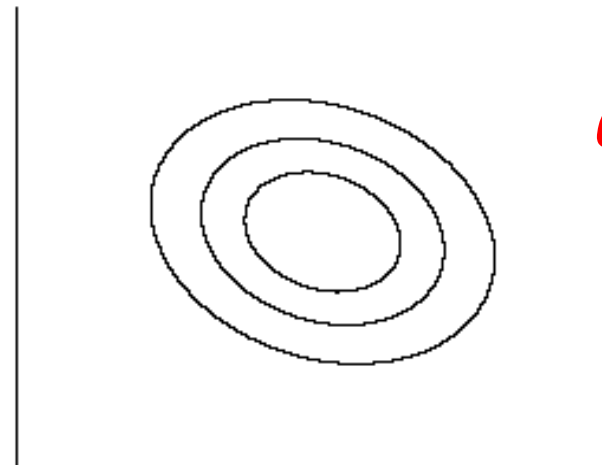
$\text{Cov}(x_1, x_2)=0, \text{Var}(x_1)>\text{Var}(x_2)$



$\text{Cov}(x_1, x_2)>0 \quad \sigma_{12}>0$



$\text{Cov}(x_1, x_2)<0 \quad \sigma_{12}<0$



explain

Multivariate Normal Distribution

- In the multivariate case, small $|\Sigma|$ means samples are close to the mean vector μ
- Σ is a symmetric positive definite matrix, or the multivariate way of saying that $\text{Var}(X) > 0$
- If Σ is not a symmetric positive definite matrix means
 - ▣ There is linear dependency between the dimensions
 - ▣ One of the dimension has variance 0
 - ▣ In such a case, we need to reduce the “dimension” (say via PCA or other techniques, which we will go over in the next chapter)

Multivariate Normal Distribution: **Naïve case**

- If x_i are independent (or $\text{COV}(X_i, X_j) = 0, \text{Var}(X_i) = \sigma_i^2, \forall i$)
- Off diagonals of Σ are 0.

$$p(\mathbf{x}) = \prod_{i=1}^d p_i(x_i) = \frac{1}{(2\pi)^{d/2} \prod_{i=1}^d \sigma_i} \exp \left[-\frac{1}{2} \sum_{i=1}^d \left(\frac{x_i - \mu_i}{\sigma_i} \right)^2 \right]$$

- Mahalanobis distance reduces to weighted (by $1/\sigma_i$)
- If variances are also equal, reduces to Euclidean distance
- We can put this $p(\mathbf{x})$ in our discriminant function

Multivariate Normal Distribution: **Property**

- Let: $\mathbf{x} \sim \mathcal{N}_d(\boldsymbol{\mu}, \Sigma)$ and $\mathbf{w} \in \mathbb{R}^d$
- We have: $\mathbf{w}^T \mathbf{x} = w_1 x_1 + w_2 x_2 + \cdots + w_d x_d \sim \mathcal{N}(\mathbf{w}^T \boldsymbol{\mu}, \mathbf{w}^T \Sigma \mathbf{w})$
- The mean and variance can be shown as:

$$\begin{aligned} E[\mathbf{w}^T \mathbf{x}] &= \mathbf{w}^T E[\mathbf{x}] = \mathbf{w}^T \boldsymbol{\mu} \\ \text{Var}(\mathbf{w}^T \mathbf{x}) &= E[(\mathbf{w}^T \mathbf{x} - \mathbf{w}^T \boldsymbol{\mu})^2] = E[(\mathbf{w}^T \mathbf{x} - \mathbf{w}^T \boldsymbol{\mu})(\mathbf{w}^T \mathbf{x} - \mathbf{w}^T \boldsymbol{\mu})] \\ &= E[\mathbf{w}^T (\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^T \mathbf{w}] = \mathbf{w}^T E[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^T] \mathbf{w} \\ &= \mathbf{w}^T \Sigma \mathbf{w} \end{aligned}$$

Show derivation

- **Result:** the projection of a d -dimensional normal on a vector $\mathbf{w}$ is *univariate normal*
- **Generalization:** project of a d -dimensional normal on $\mathbf{W}$ which is a $d \times k$ matrix with rank $k < d$, then

$\mathbf{W}^T \mathbf{x}$ is k -variate normal:

Multivariate Parametric Classification

- When $\mathbf{x} \in \mathcal{X}^d$
- If conditional densities are $p(\mathbf{x}|C_i) \sim N_d(\mu_i, \Sigma_i)$

$$p(\mathbf{x}|C_i) = \frac{1}{(2\pi)^{d/2} |\Sigma_i|^{1/2}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_i)^T \boldsymbol{\Sigma}_i^{-1} (\mathbf{x} - \boldsymbol{\mu}_i) \right]$$

- **Application:** predict the type of car a customer is interested in
- Different cars = classes
- Observable data of customers: $\mathbf{x}$
- Vector of mean age/income of customers who buy car type i : $\boldsymbol{\mu}_i$
- $\boldsymbol{\Sigma}_i$: Covariance matrix:
 - σ_{i1}^2 : variance of age of type i
 - σ_{i2}^2 : variance of income of type i
 - σ_{i12} : covariance of age & income in the group of customers who buy type i

Multivariate Parametric Classification

- Discriminant functions and assuming $p(\mathbf{x}|C_i) \sim \mathcal{N}_d(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$

$$g_i(\mathbf{x}) = \log p(\mathbf{x}|C_i) + \log P(C_i)$$

$$g_i(\mathbf{x}) = -\frac{d}{2} \log 2\pi - \frac{1}{2} \log |\boldsymbol{\Sigma}_i| - \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_i)^T \boldsymbol{\Sigma}_i^{-1} (\mathbf{x} - \boldsymbol{\mu}_i) + \log P(C_i)$$

- Training sample for $K \geq 2$ classes:

$$\mathcal{X} = \{\mathbf{x}^t, \mathbf{r}^t\}, \text{ where } r_i^t = 1 \text{ if } \mathbf{x}^t \in C_i \text{ and } 0 \text{ otherwise}$$

- How can we do the parametric estimate?

Estimation of Parameters

$$\hat{P}(C_i) = \frac{\sum_t r_i^t}{N}$$

$$\mathbf{m}_i = \frac{\sum_t r_i^t \mathbf{x}^t}{\sum_t r_i^t} \quad \text{Note: this is a (dx1) vector}$$

$$\mathbf{S}_i = \frac{\sum_t r_i^t (\mathbf{x}^t - \mathbf{m}_i)(\mathbf{x}^t - \mathbf{m}_i)^T}{\sum_t r_i^t} \quad \text{Note: this is a (dxd) matrix}$$

Plug the above expressions in

$$g_i(\mathbf{x}) = -\frac{1}{2} \log |\mathbf{S}_i| - \frac{1}{2} (\mathbf{x} - \mathbf{m}_i)^T \mathbf{S}_i^{-1} (\mathbf{x} - \mathbf{m}_i) + \log \hat{P}(C_i)$$

Expand it

$$g_i(\mathbf{x}) = -\frac{1}{2} \log |\mathbf{S}_i| - \frac{1}{2} \left(\mathbf{x}^T \mathbf{S}_i^{-1} \mathbf{x} - 2\mathbf{x}^T \mathbf{S}_i^{-1} \mathbf{m}_i + \mathbf{m}_i^T \mathbf{S}_i^{-1} \mathbf{m}_i \right) + \log \hat{P}(C_i)$$

Different S_i

- The above discriminant is a “Quadratic discriminant”

$$g_i(\mathbf{x}) = \mathbf{x}^T \mathbf{W}_i \mathbf{x} + \mathbf{w}_i^T \mathbf{x} + w_{i0} \quad \text{Why quadratic?}$$

where

$$\mathbf{W}_i = -\frac{1}{2} \mathbf{S}_i^{-1}$$

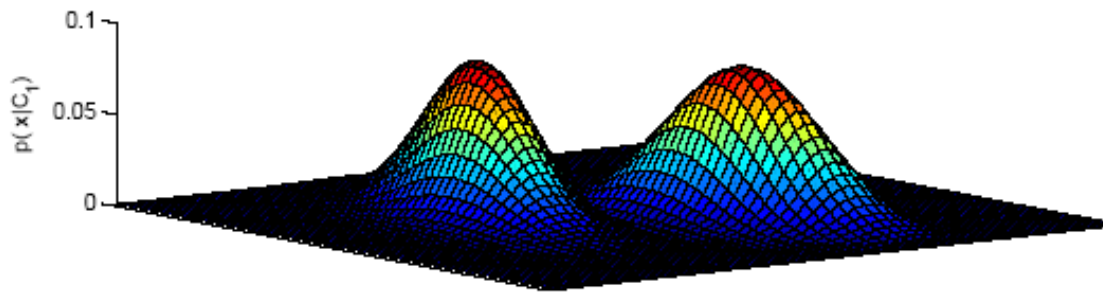
Number of parameters to estimate:

$$\mathbf{w}_i = \mathbf{S}_i^{-1} \mathbf{m}_i$$

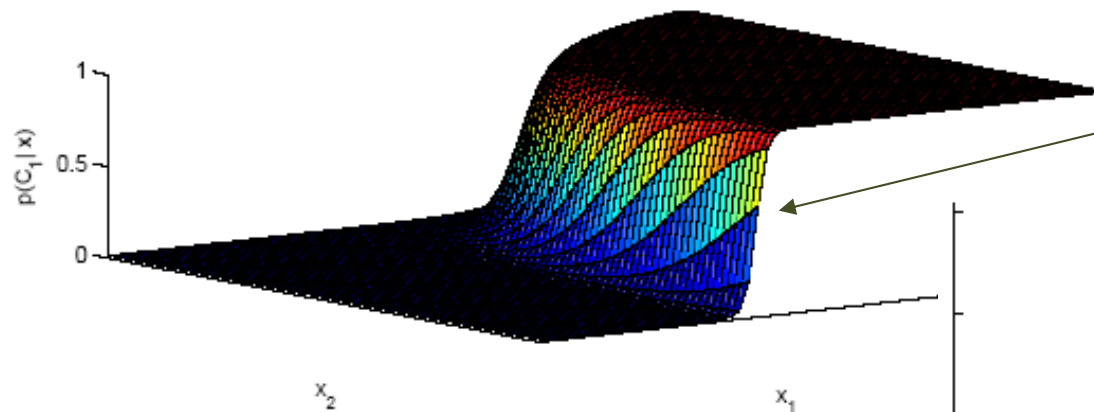
$$w_{i0} = -\frac{1}{2} \mathbf{m}_i^T \mathbf{S}_i^{-1} \mathbf{m}_i - \frac{1}{2} \log |\mathbf{S}_i| + \log \hat{P}(C_i)$$

- $K*d$ for means
- $K*d(d+1)/2$ for covariance matrices

- S_i may be singular and inverse may not exist
- $|S_i|$ may be nonzero but very small, makes the process unstable
- If small # of samples, need to reduce dimension d (**CHAPTER 6**)

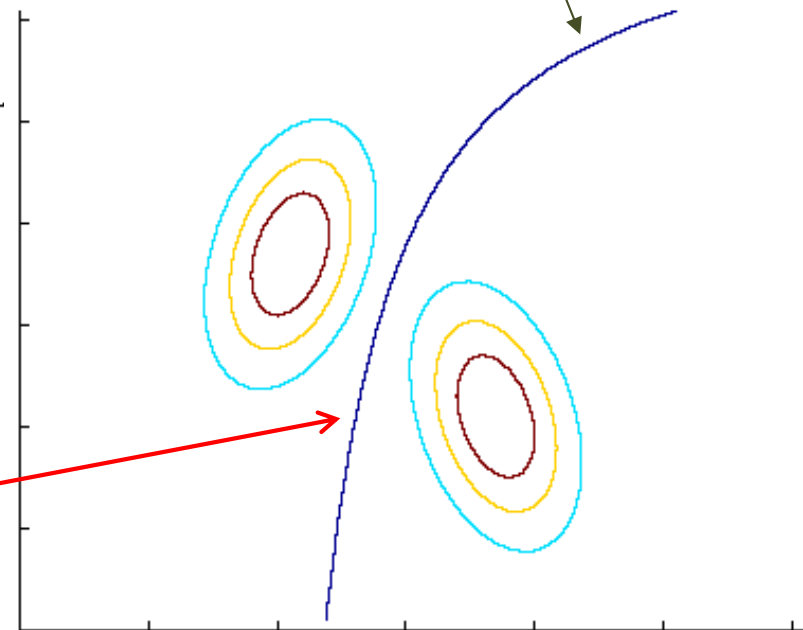


likelihoods



posterior for C_1

discriminant:
 $P(C_1 | \mathbf{x}) = 0.5$



Quadratic discriminant

Case 1: Common Covariance Matrix $\mathbf{S}$

- Shared common sample covariance $\mathbf{S}$

$$\mathbf{S} = \sum_i \hat{P}(C_i) \mathbf{S}_i$$

- Discriminant reduces to Common $\mathbf{x}^T \mathbf{S}^{-1} \mathbf{x} \quad \forall C_i$, can be ignored

$$g_i(\mathbf{x}) = -\frac{1}{2} (\mathbf{x} - \mathbf{m}_i)^T \mathbf{S}^{-1} (\mathbf{x} - \mathbf{m}_i) + \log \hat{P}(C_i)$$

which becomes a **linear discriminant**

$$g_i(\mathbf{x}) = \mathbf{w}_i^T \mathbf{x} + w_{i0}$$

where

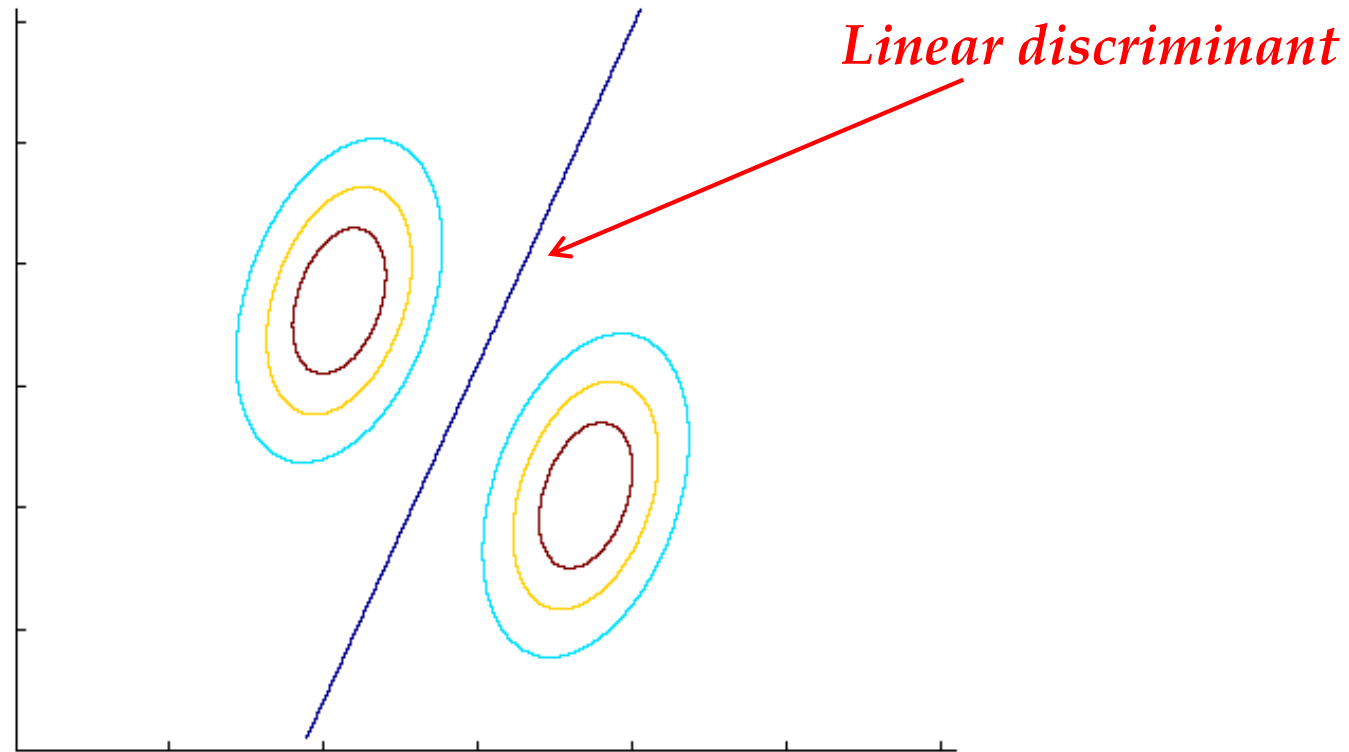
$$\mathbf{w}_i = \mathbf{S}^{-1} \mathbf{m}_i$$

$$w_{i0} = -\frac{1}{2} \mathbf{m}_i^T \mathbf{S}^{-1} \mathbf{m}_i + \log \hat{P}(C_i)$$

Number of parameters to estimate:

- $K*d$ for means
- $d(d+1)/2$ for covariance matrices

Common Covariance Matrix S



If the priors are equal, the “optimal decision” is to assign input to the class whose mean’s Mahalanobis distance to the input is the smallest.

Case 2: Diagonal S

- If we assume all off-diagonals of the covariance matrix are zeros (or when $x_i, i = 1, \dots, d$ are independent)
- This is the **“Naïve Bayes’ classifier”**: Σ is diagonal, $p(x_i|C_i)$ are univariate Gaussian, S and its inverse are diagonal

$p(\mathbf{x}|C_i) = \prod_d p(x_d|C_i)$, or Naive Bayes’ assumption

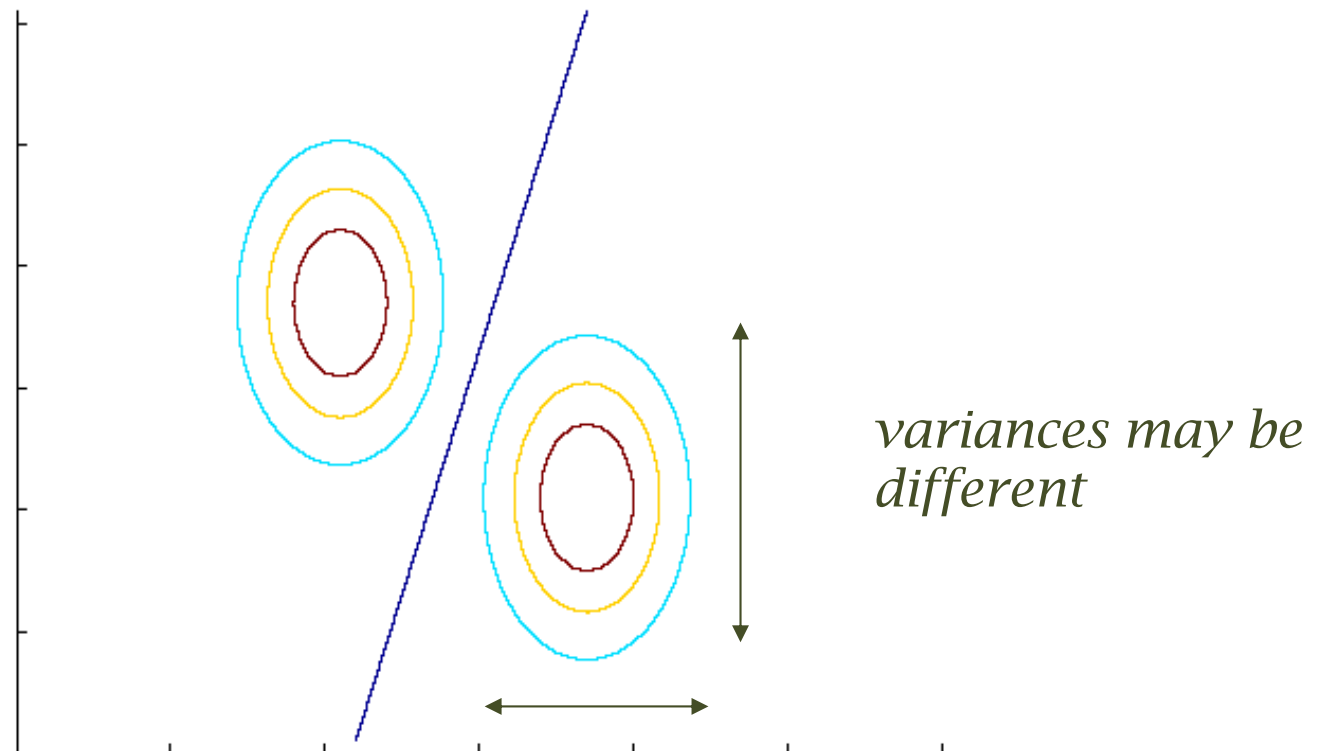
$$g_i(\mathbf{x}) = -\frac{1}{2} \sum_{j=1}^d \left(\frac{x_j^t - m_{ij}}{s_j} \right)^2 + \log \hat{P}(C_i)$$

Classify based on weighted Euclidean distance (in s_j units) to the nearest mean

Number of parameters to estimate:

- $K*d$ for means
- d for covariance matrices

Diagonal S



Diagonal S, equal variances

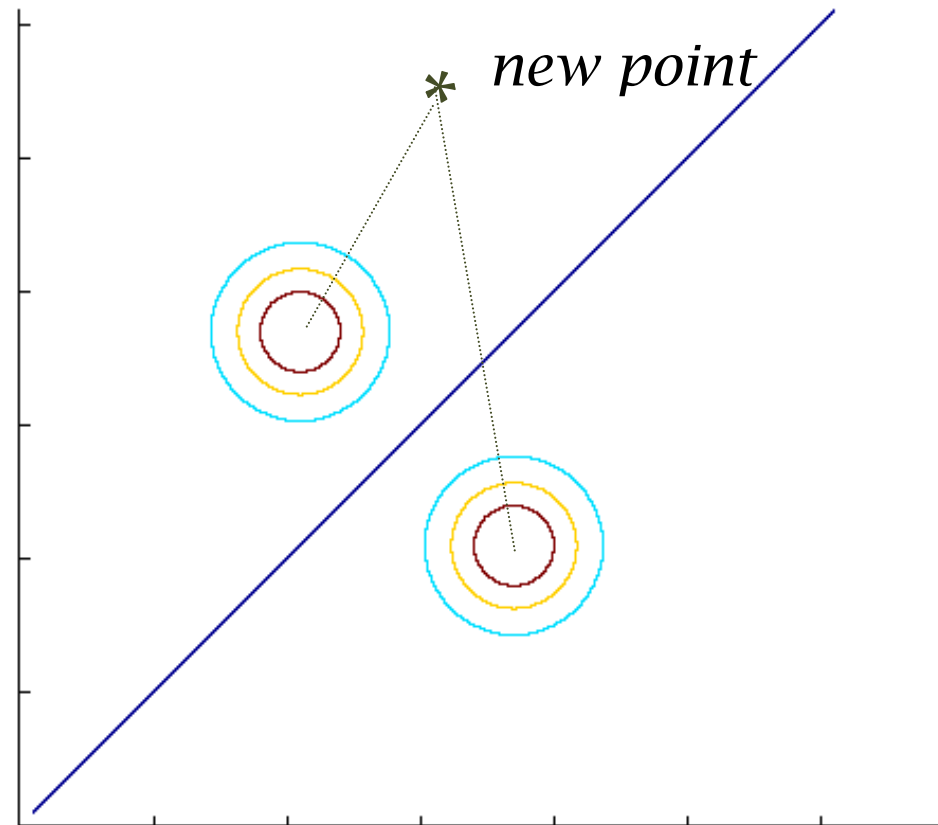
- When **all variances are equal**, $|S| = s^{2d}$, $S^{-1} = (1/s^2)$, Mahalanobis distance reduces to the *Euclidean distance*
- **Discriminant function:**
$$g_i(\mathbf{x}) = -\frac{\|\mathbf{x} - \mathbf{m}_i\|^2}{2s^2} + \log \hat{P}(C_i) = -\frac{1}{2s^2} \sum_{j=1}^d (x_j^t - m_{ij})^2 + \log \hat{P}(C_i)$$
- When the priors are the same ($\hat{P}(C_i)$), it becomes the **nearest mean classifier: Classify based on Euclidean distance to the nearest mean**
- Each mean can be considered a prototype or template and this **is template matching**

Number of parameters to estimate:

- $K*d$ for means
- 1 for covariance matrices

$$\begin{aligned} g_i(\mathbf{x}) &= -\|\mathbf{x} - \mathbf{m}_i\|^2 = -(\mathbf{x} - \mathbf{m}_i)^T (\mathbf{x} - \mathbf{m}_i) \\ &= -(\mathbf{x}^T \mathbf{x} - 2\mathbf{m}_i^T \mathbf{x} + \mathbf{m}_i^T \mathbf{m}_i) \end{aligned}$$

Diagonal S , equal variances

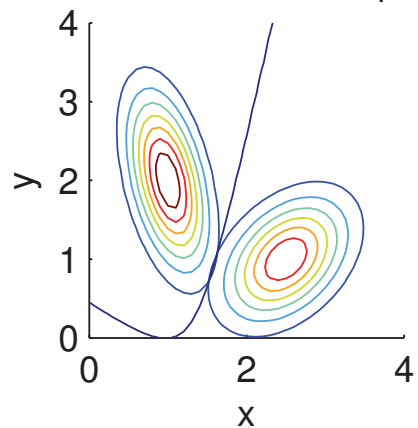


Tuning Complexity: Model Selection

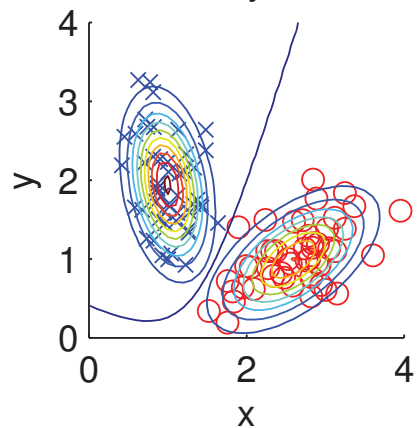
<i>Assumption</i>	<i>Covariance matrix</i>	<i>No of parameters in $\mathbf{S}_i$</i>
Shared, Hyperspheric	$\mathbf{S}_i = \mathbf{S} = s^2 \mathbf{I}$	1
Shared, Axis-aligned	$\mathbf{S}_i = \mathbf{S}$, with $s_{ii} = 0$	d
Shared, Hyperellipsoidal	$\mathbf{S}_i = \mathbf{S}$	$d(d+1)/2$
Different, Hyperellipsoidal	$\mathbf{S}_i$	$K d(d+1)/2$

- As we increase complexity (less restricted $\mathbf{S}$), bias decreases and variance increases (**bias/variance dilemma**)
- Assume simple models (allow some bias) to control variance (regularization)

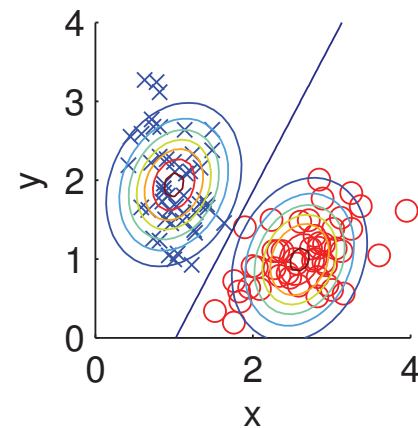
Population likelihoods and posteriors



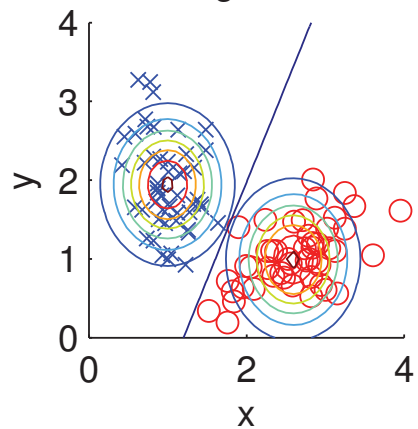
Arbitrary covar.



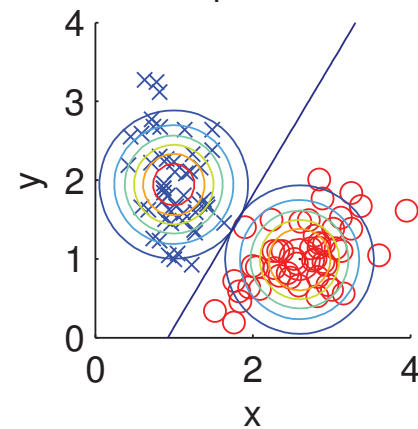
Shared covar.



Diag. covar.



Equal var.



Discrete Features

- **Binary** features: $p_{ij} \equiv p(x_j = 1 | C_i)$ **Read book on application**
if x_i are **independent** (Naive Bayes')

$$p(\mathbf{x} | C_i) = \prod_{j=1}^d p_{ij}^{x_j} (1 - p_{ij})^{(1-x_j)}$$

the discriminant is linear

$$\begin{aligned} g_i(\mathbf{x}) &= \log p(\mathbf{x} | C_i) + \log P(C_i) \\ &= \sum_j \left[x_j \log p_{ij} + (1 - x_j) \log(1 - p_{ij}) \right] + \log P(C_i) \end{aligned}$$

Estimated parameters

$$\hat{p}_{ij} = \frac{\sum_t x_j^t r_i^t}{\sum_t r_i^t}$$

Discrete Features

- **Multinomial** (1-of- n_j) features: $x_j \in \{v_1, v_2, \dots, v_{n_j}\}$

- Define new variables:

$$z_{jk}^t = \begin{cases} 1 & \text{if } x_j^t = v_k \\ 0 & \text{otherwise} \end{cases}$$

- **Let** p_{ijk} denote the probability that x_j belonging to C_i takes value v_k :

$$p_{ijk} \equiv p(z_{jk} = 1 | C_i) = p(x_j = v_k | C_i) \quad \sum_{k=1}^{n_j} p_{ijk} = 1$$

- If all attributes are **independent**, we have

$$p(\mathbf{x} | C_i) = \prod_{j=1}^d \prod_{k=1}^{n_j} p_{ijk}^{z_{jk}}$$

$$g_i(\mathbf{x}) = \sum_j \sum_k z_{jk} \log p_{ijk} + \log P(C_i)$$

- **Maximum likelihood estimator:** $\hat{p}_{ijk} = \frac{\sum_t z_{jk}^t r_i^t}{\sum_t r_i^t}$

Multivariate Regression

- **Multivariate linear model:** numeric output r is a weighted sum of all d variables, $x_1, \dots, x_d$, and noise.

$$r^t = g(\mathbf{x}^t | w_0, w_1, \dots, w_d) + \epsilon = w_0 + w_1 x_1^t + w_2 x_2^t + \dots + w_d x_d^t + \epsilon$$

- When noise is normal with 0 mean and constant variance, maximize the likelihood is to minimize the sum square error (*page 33 for slide in Chapter 4*):

$$E(w_0, w_1, \dots, w_d | \mathcal{X}) = \frac{1}{2} \sum_t (r^t - w_0 - w_1 x_1^t - w_2 x_2^t - \dots - w_d x_d^t)^2$$

- Taking the derivative with respect to $w_j, j = 0, \dots, d$

$$\begin{aligned} \sum_t r^t &= Nw_0 + w_1 \sum_t x_1^t + w_2 \sum_t x_2^t + \dots + w_d \sum_t x_d^t \\ \sum_t x_1^t r^t &= w_0 \sum_t x_1^t + w_1 \sum_t (x_1^t)^2 + w_2 \sum_t x_1^t x_2^t + \dots + w_d \sum_t x_1^t x_d^t \\ \sum_t x_2^t r^t &= w_0 \sum_t x_2^t + w_1 \sum_t x_1^t x_2^t + w_2 \sum_t (x_2^t)^2 + \dots + w_d \sum_t x_2^t x_d^t \\ &\vdots \\ \sum_t x_d^t r^t &= w_0 \sum_t x_d^t + w_1 \sum_t x_1^t x_d^t + w_2 \sum_t x_2^t x_d^t + \dots + w_d \sum_t (x_d^t)^2 \end{aligned}$$

Multivariate Regression

- Define vectors and matrix:

$$\mathbf{X} = \begin{bmatrix} 1 & x_1^1 & x_2^1 & \cdots & x_d^1 \\ 1 & x_1^2 & x_2^2 & \cdots & x_d^2 \\ \vdots & & & & \\ 1 & x_1^N & x_2^N & \cdots & x_d^N \end{bmatrix}, \mathbf{w} = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_d \end{bmatrix}, \mathbf{r} = \begin{bmatrix} r^1 \\ r^2 \\ \vdots \\ r^N \end{bmatrix}$$

- **Normal equation:** $\mathbf{X}^T \mathbf{X} \mathbf{w} = \mathbf{X}^T \mathbf{r}$, or solve for: $\mathbf{w} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{r}$
- Same as we did for polynomial regression using one input when we define: $x_1 = x, x_2 = x^2, \dots, x_k = x^k$
- This is known as the **multivariate polynomial regression**
- We can use this method if d is small
- If w_i is **+** (**-**), it has a **positive** (**negative**) effect on the output
- If all x_i are in the same range, w_i indicates which features are important

Conclusion

- We extend the parametric method in Chapter 4 (single feature) to general d -features
- Consider multivariate distributions: Gaussian and multinomial d -feature distribution
- Multivariate Gaussian: features not necessary independent.
 - ▣ Depending on my assumptions, we have different type of classifier (from quadratic to linear to nearest distance)
- Multinomial distribution: assume independence in features
- **Question:** *What are the short-comings of this method?*